

Business Problem Statement

Customer Churn Analysis — Subscription Streaming Business

1. Business Background

PlayWave is a subscription-based digital streaming platform operating across India and Nepal, offering Basic, Standard, and Premium subscription tiers on both Monthly and Annual contracts. The company acquires customers through paid marketing, organic search, and referral programs. Over the past several quarters, the Subscriber Retention team has observed fluctuations in the number of customers cancelling their subscriptions, prompting a need to better understand the underlying drivers of customer churn and its financial impact on the business.

2. Problem Statement

The company currently lacks a consolidated, data-driven view of why customers cancel their subscriptions, which customer segments are most at risk of churning, and how much recurring revenue is exposed to that risk. Without this visibility, retention campaigns, customer support resourcing, and pricing/plan strategy are being decided without clear evidence, resulting in inefficient use of marketing and retention budgets and continued preventable revenue loss.

3. Business Objectives

- Quantify the overall customer churn and retention rate across the subscriber base.
- Identify which subscription plans, contract types, and geographies have the highest churn.
- Estimate the monthly recurring revenue (MRR) that is at risk due to churn, segmented by plan.
- Determine the primary reasons customers cite for cancelling their subscription.
- Evaluate whether customer support experience (CSAT, escalations, complaint volume) is correlated with churn.
- Segment customers into churn-risk tiers (Low / Medium / High) to prioritize retention outreach.
- Track how churn has trended over time to detect seasonal or campaign-driven spikes.

4. Key Business Questions

- What percentage of customers have churned, and what is the current retention rate?
- Which plan type (Basic, Standard, Premium) and contract type (Monthly, Annual) has the highest churn rate?
- How much monthly recurring revenue (MRR) and customer lifetime value (CLTV) is being lost to churn?
- What are the top reasons customers give for cancelling their subscription?
- Is there a relationship between support ticket escalations / CSAT scores and the likelihood of churn?
- Which customer segments fall into the High-Risk churn category, and how much revenue do they represent?
- How has the monthly/quarterly churn trend evolved, and are there any noticeable spikes?

5. Scope & Available Data

The analysis is based on three internal data sources maintained by the company, joined on a common Customer ID to build a unified view of each subscriber's profile, billing relationship, and support history:

Data Source	Key Fields
Customer Master	Customer ID, name, country, state, gender, date of birth, interests
Subscription	Plan type, contract type, monthly charges, CLTV, subscription/cancellation dates, cancellation reason, churn score
Customer Support	Complaint date, escalation flag, CSAT score, comments

6. Expected Deliverables

- Overall churn rate, retention rate, and average churn score across the customer base.
- Revenue KPIs: Monthly Recurring Revenue (MRR), Average Revenue Per User (ARPU), and Average CLTV.
- Churn rate breakdowns by plan type, contract type, and state/region.
- A ranked summary of the most common cancellation reasons.
- Support-experience metrics (average CSAT, escalation rate) and their correlation with churn.
- Customer segmentation into Low / Medium / High churn-risk tiers, with revenue-at-risk quantified per tier and per plan.
- A monthly/quarterly churn trend visualization to identify patterns over time.
- A pivot-table view summarizing revenue at risk by plan type and churn-risk tier, to help prioritize retention efforts.

7. Success Criteria

This analysis will be considered successful if it provides the Retention and Customer Success teams with clear, quantified answers to the key business questions above, enabling them to prioritize retention campaigns toward the highest-revenue-risk segments, and to make an evidence-based case for any changes to plan pricing, contract incentives, or customer support processes.

PlayWave is a fictional company created for this portfolio project. Any resemblance to a real company is coincidental.